MAT A31

Week 3 Lectures: Notes

Textbook: Laura Taalman, Peter Kohn**:** CALCULUS, *Single Variable,1e*

Logic and Mathematical Thinking

A mathematical proof is a logical argument. Every theorem in this course can be proved with the use of previous theorems or definitions.

3.1 Language of proofs.

Quantifiers.

In the process of proving we'll need to state if a property
"true all the time", "some of the time", "at least once", "none of the time".

Logical *quantifiers* make such statements mathematically precise.

"For All"

Symbol $\forall$ **"for all"**

"***There Exists***"

Symbol $\exists$ - "**there exists**"

Implications.

The statements in the form "if A then B" ($A \Rightarrow B$) are called *implications.*
' $A \Rightarrow B$ ' means that whenever statement A is true, statement B is also true.
Here A is the *hypothesis* and B is the *conclusion*.

Ex: a) $2x-1>0 \Rightarrow x>0$

b) If ***a*** and ***b*** are both odd numbers *then* ***a*** + ***b*** is even number

c) If two angles are congruent *then* they have the same measurements

The statement $B \Rightarrow A$ is the *converse of the implication* $A \Rightarrow B$.

a)

b)

c)

The statement 'Not A $\Rightarrow$ 'Not B' is the *inverse of the implication* $A \Rightarrow B$.
$\neg A \Rightarrow \neg B$

a)

b)

c)

The statement 'Not B ' $\Rightarrow$ 'Not A' is the *contrapositive of the implication* $A \Rightarrow B$

$\neg B \Rightarrow \neg A$

a)

b)

c)

If the implication and its converse are both true, then we can write a *two-way implication* $A \Leftrightarrow B$

$A \Leftrightarrow B$ is reads as "A equivalent to B' or: " A if and only if B" or: "A iff B"

a)

b)

The definition based on contrapositive of the implication is its logical equivalent.

Counterexamples.

Counterexamples are examples that prove that a statement is false.

- 'for all real *x*, $x = 2k+1,\ where\ \ k$ is integer ' is FALSE
- (counterexample:)

- ' for all *x* , if *x* is positive, then *x* is odd' is FALSE
  (counterexample:)

3.2 Mathematical proofs.

a) Calculational proof.

Example 1. Prove that $x^2 - y^2 = (x - y)(x + y)$ for all real numbers.

PROOF:

QED.

b) Direct Proof

Example 2: Prove that the sum of two odd numbers is even.

<u>Given</u>:

<u>Show:</u>

PROOF:

***c) A proof by contradiction*.**

Example 3: Let $f(x) = 2x + 5$. Prove that $f(x)$ is one-to –one function.

Given:

Show:

PROOF:

Mathematical notations.

$\forall$ - "for all"
$\ni$, st - "such that"
$\exists$ - "there exist"
$\in$ - " belongs to" "is in …"
$\Rightarrow$ - "implies"
$\Leftrightarrow$ - "equivalent", "iff"
$\notin$ - " is not in.."

$|$ - "such that " (in sets)

Numbers:
R – set of real numbers
I – set of irrational numbers
Q - set of rational numbers
Z – set of integers
N – natural numbers

(nonnegative integers and 0)

EXAMPLES

1. Prove that if x is irrational number and r is a rational number, then the sum $r+x$ is an irrational number.

Given:

Show:

PROOF:

2. Prove that for any real numbers a and b $|a-b|=|b-a|$

3. Prove the distance formula $D=\sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$

PROOF